

EDUCATION

MSc Computer Engineering	Vienna University of Technology	Oct. 2022 - Nov. 2025
Bridging Program Embedded Systems	TU Delft	Sept. 2020 - Jul. 2022
BSc Computer Science	Amsterdam University of Applied Sciences	Sept. 2016 - Jul. 2020

PUBLICATIONS & AWARDS

- J. He, **V. T. W. Kenbeek**, Z. Yang, M. Qu, E. Bartocci, D. Ničković, and R. Grosu, “TD-Interpreter: Enhancing the Understanding of Timing Diagrams with Visual-Language Learning”, *arXiv preprint arXiv:2507.16844*, 2025. doi: 10.48550/arXiv.2507.16844.
- Young Academics 2024 Award from ESBS-Austria for an outstanding master’s thesis in electronics and AI for chips

WORK EXPERIENCE

IoT Research Engineer	Research Studios Austria FG	Feb 2026 - Present
- Contributed to proposal writing for a Horizon Chips JU project, leading two use cases and coordinating partner contributions during proposal preparation. - Presented current research activities during the Lange Nacht der Forschung (Long Night of Research). - Designed and implemented a hydroponics control system by integrating an LLM (Qwen3-4B) using llama.cpp to generate control actions based on the system state. - Optimized the hydroponic system’s model by quantizing the model using OpenVINO for efficient inference on an industrial PC with limited resources. - Implemented a RAG pipeline using LangChain and Chroma to retrieve requirements from engineering standards.		
C Developer	Open-Xchange	Apr 2024 - January 2026
- Developed and maintained the Dovecot email server stack in C. - Integrated Azure Blob Storage into Dovecot. - Applied test-driven development using pytest.		
RISC-V Embedded Engineer	OQmented	June 2023 - December 2023
- Researched and evaluated open-source RISC-V microcontroller solutions for AR glasses hardware. - Configured toolchains and compilers for the X-HEEP platform. - Ported the X-HEEP platform from Xilinx to Lattice. - Conducted requirements analysis from product specifications and engineering meetings.		
Embedded Software Engineer	Inventeers R&D	June 2021 - August 2022
- Designed, implemented and tested a bootloader in C and Assembly with over-the-air firmware update capability over LoRa for resource-constrained IoT devices. - Integrated USB functionality using TinyUSB on SAM D microcontrollers. - Developed and maintained firmware for various SAM D devices in C++20 across multiple product lines.		

SKILLS

<u>Programming Languages:</u>		<u>Technologies:</u>	
Proficient	C, C++, Python, JavaScript	AI	PyTorch, RAG, LangChain, Ollama, llama.cpp, OpenVINO
Intermediate	Assembly, Java, C#	Protocols	MQTT, CoAP, LoRaWAN, USB, I2C, UART, SPI, BLE
Beginner	Verilog, VHDL, CUDA	Databases	MySQL, PostgreSQL, InfluxDB, Vector Databases
		Other	Linux, Git, ROS, Docker, Grafana

Human Languages: Dutch (native), English (C1 - Cambridge Advanced Certificate), German (basic)